



# Occurrence and cross contamination of *Escherichia albertii* in retail chicken outlets in Bangladesh

Jayedul Hassan<sup>a</sup>, Kishor Sosmith Utsho<sup>a</sup>, Susmita Karmakar<sup>a</sup>, Md. Wohab Ali<sup>a</sup>,  
Sharda Prasad Awasthi<sup>b,c,d</sup>, Chiharu Uyama<sup>e</sup>, Noritoshi Hatanaka<sup>b,c,d,e</sup>, Shinji Yamasaki<sup>b,c,d,e</sup>,  
Atsushi Hinenoya<sup>b,c,d,e,\*</sup>

<sup>a</sup> Department of Microbiology and Hygiene, Bangladesh Agricultural University, Mymensingh, Bangladesh

<sup>b</sup> Graduate School of Veterinary Science, Osaka Metropolitan University, Osaka, Japan

<sup>c</sup> Asian Health Science Institute, Osaka Metropolitan University, Osaka, Japan

<sup>d</sup> Osaka International Research Center for Infectious Diseases, Osaka Metropolitan University, Osaka, Japan

<sup>e</sup> School of Life and Environmental Sciences, Osaka Prefecture University, Osaka, Japan

## ARTICLE INFO

### Keywords:

*Escherichia albertii*

Retail chicken

Bangladesh

Cross contamination

Antimicrobial resistance

Multidrug resistance

## ABSTRACT

*Escherichia albertii* is an emerging zoonotic pathogen linked to human gastrointestinal illnesses, with poultry meats being considered as a key source of human infections. However, there is little information regarding the distribution and characteristics of this bacterium in Bangladesh. This study investigated the occurrence, antimicrobial resistance, and virulence of *E. albertii* in chicken meats from retail outlets in Bangladesh. We collected samples from 61 dressed chickens across 17 retail shops from 4 upazilas, along with swabs from cloaca, processing utensils, and worker hands. Detection of *E. albertii* by species-specific PCR revealed substantial occurrence of *E. albertii* in retail chicken meat (63.9 %), cloaca (71.4 %), human hand (45.5 %), bleeding cone (13.3 %) and blade (10 %). Almost all the *E. albertii* isolates (94.4 %) exhibited resistance to at least one of the tested antimicrobials, among which 50 % were multidrug resistant, including resistance to clinically relevant antimicrobials such as tetracycline, ampicillin, gentamicin, kanamycin, nalidixic acid and ciprofloxacin. Whole genome sequencing analysis identified the presence of corresponding antimicrobial resistance genes and critical virulence genes (*eae*, *Eacdt*). Notably, although wgSNP-based phylogenetic analysis showed the genomic diversity of the isolates, some of the isolates from the same shop displayed clonal relationships among meats, cloacal swabs, and human hand swabs, indicating contamination during processing. These findings highlight the public health risk posed by *E. albertii* in retail poultry, underlining the poultry's role as a potential vector for zoonotic transmission and the need for improved biosecurity and antimicrobial management practices in poultry production.

## 1. Introduction

*Escherichia albertii* is an emerging zoonotic enteropathogen causing gastrointestinal diseases in humans, such as watery diarrhea, abdominal pain, vomiting and fever (Bhatt et al., 2019; Masuda et al., 2020; Oaks et al., 2010). This bacterium was first isolated from a diarrheic child in Bangladesh (Albert et al., 1991), but was initially identified as *Hafnia alvei* using the API 20E biochemical identification strip. Subsequent extensive analyses of this *H. alvei* isolate proposed a new member of Genus *Escherichia*, *E. albertii* (Huys et al., 2003). In addition to an increase in clinical reports worldwide, several foodborne outbreaks caused by *E. albertii* have occurred, especially in Japan (Kashio et al., 2020;

Masuda et al., 2020; Ooka et al., 2013; Tokoi et al., 2018), and consequently, this bacterium has garnered increasing attention with respect to public health globally. *E. albertii* is a Gram-negative, non-spore forming, non-motile (in conventional bacteriological tests), facultative anaerobic bacillus. *E. albertii* carries the locus of enterocyte effacement (LEE) pathogenicity island encoding type three secretion system and *eae* gene-encoded adhesin called intimin, by which this bacterium can form attaching and effacing lesion on intestinal epithelium. *E. albertii* also express CDTs encoded by *Eacdt* (almost ubiquitously present) and *Eccdt-I* (less frequently present), which could be associated with virulence and persistent colonization (Scuron et al., 2016). Certain strains also produce Shiga toxin 2 (Stx2a or Stx2f) (Brandal et al., 2015; Hinenoya et al.,

\* Corresponding author at: Graduate School of Veterinary Science, Osaka Metropolitan University, 1-58, Rinku Ourai-kita, Izumisano, Osaka 598-8531, Japan.  
E-mail address: [hinenoya@omu.ac.jp](mailto:hinenoya@omu.ac.jp) (A. Hinenoya).

2017a; Murakami et al., 2014), which is a primary virulence factor of enterohemorrhagic *E. coli*. Due to such similar phenotypic and genotypic feature with other bacteria belonging to the order Enterobacterales, *E. albertii* has often been misidentified as *H. alvei*, *Shigella boydii*, enteropathogenic *E. coli*, EHEC and CDT-II-producing *E. coli* (Hinenoya et al., 2019a; Hinenoya et al., 2017b; Hyma et al., 2005; Ooka et al., 2012), suggesting the underestimation of *E. albertii* infections in humans.

Many epidemiological studies have explored the environmental distribution of *E. albertii*, identifying animal reservoirs and potential human infection routes. Wildlife surveys reported this bacterium in several animals, such as raccoons, badgers, martens, foxes, bats, dogs and cats (Gordon, 2011; Hinenoya et al., 2020a; Hinenoya et al., 2022b; Naka et al., 2024; Naka et al., 2022; Ooka et al., 2012), suggesting that these animals may serve as reservoirs or carriers. Notably, raccoons are considered as the significant natural reservoir of *E. albertii* (Hinenoya et al., 2020a; Xu et al., 2023). Additionally, wild birds have also been shown to frequently carry this bacterium in several countries (Barmettler et al., 2022; Hinenoya et al., 2022a; Liu et al., 2022b; Naka et al., 2022). Moreover, among livestock, *E. albertii* has been predominantly detected in poultry (Barmettler et al., 2023; Gordon, 2011; Hinenoya et al., 2021; Oaks et al., 2010; Wang et al., 2022) and their meats (Asoshima et al., 2015; Gupta et al., 2023; Konno et al., 2021; Lindsey et al., 2015; Maeda et al., 2015), with a higher occurrence than in other meats (beef, mutton and pork) (Wang et al., 2016). Thus, poultry are considered as an important reservoir for human *E. albertii* infections.

The poultry industry is a major contributor to human nutrition across the world, providing affordable and accessible protein sources. However, the poultry supply chain faces significant challenges, particularly regarding the emergence and spreading of antimicrobial resistance (AMR) in bacteria, including the zoonotic pathogens (Ferdous et al., 2023; Samad et al., 2023). AMR and multidrug-resistant *E. albertii* have also been identified in isolates from chickens (Hinenoya et al., 2021; Wang et al., 2022). Overcrowded and unhygienic conditions on farms facilitate the spread of infectious diseases among birds, often necessitating the extensive use of antimicrobials as prophylactics and growth promoters (Ferdous et al., 2023; Imam et al., 2023). This indiscriminate use of antimicrobials accelerates the development and selection of antimicrobial-resistant bacteria, which can enter the human food chain via contaminated meats. In developing countries, where the food cold chain is not well developed, live poultry markets and shops are common to keep meats fresh. Poultry is slaughtered just before the products are provided to consumers. Processing generally lacks strict sanitation protocols and is carried out in areas not isolated from the live birds, suggesting the high risk of microbial contamination among poultry products, food handlers, and processing tools (Chowdhury et al., 2020; Uddin et al., 2019). In Bangladesh, chickens are the most ubiquitous and abundant species of domestic livestock. A recent study has shown the presence and AMR profiles of *E. albertii* in backyard chickens (Gupta et al., 2023), but the situation of *E. albertii* in retail chicken meats from the poultry industry remains unknown. Thus, epidemiological studies are warranted to assess the occurrence, virulence, and AMR of *E. albertii* in retail chickens in Bangladesh. Such studies would help to evaluate the potential role of chicken meats as a source of human infections.

Thus, this study aimed to investigate the occurrence of *E. albertii* in chicken meats sold at the retail chicken shops in Bangladesh and characterize the isolates through their genotypic and phenotypic analyses, including whole-genome sequencing. This study will provide critical insights into the role of poultry meats in the transmission of *E. albertii* to humans and the public health risks associated with the pathogen's AMR in Bangladesh.

## 2. Materials and methods

### 2.1. Ethical approval

Verbal approval was taken from the retail shop owners and associated personnels from whom the samples were collected. The study design and methodologies were approved by the Animal Welfare and Experimentation Ethics Committee, Bangladesh Agricultural University [Approval no. AWEEC/BAU/2021 (48)].

### 2.2. Sample collection and processing

Whole dressed broiler chickens without skin ( $n = 61$ ), along with their giblets (liver, proventriculus-gizzard), were purchased from 17 retail outlets in 4 upazila (Sadar, Tarakanda, Trishal and Muktagacha) in Mymensingh district, Bangladesh between October 2021 and March 2023 (Supplementary Fig. 1). The 3 markets (K. R. market, Bangladesh Agricultural University; Kewatkhali Bazar, Ketwatkhali; Mechua Bazar, Mechua) are managed by Mymensingh city corporation, and others including Tarakanda Bazar, Trishal Bazar and Muktagacha Bazar are managed by their own municipalities. The chickens were slaughtered and dressed (removed skin and separated internal organs) at the outlets by the owners or workers according to their regular practices. The meat, liver and proventriculus-gizzard (P-G) from each chicken were put separately in plastic bags, labeled accordingly, and transported to the laboratory in ice box. At the laboratory, thigh and breast muscles were separated aseptically. Afterward, four different types of samples - thigh, breast, liver, P-G - from each chicken were subjected to microbiological analysis for *E. albertii*. Twenty-five gram of each sample was homogenized in 225 mL buffered peptone water (HiMedia Laboratories Private Ltd., Mumbai, Maharashtra, India), and incubated overnight (~20 h) at 37 °C for enrichment. After November 2022, cloacal swabs ( $n = 35$ ) from the chickens before slaughtering, hand swabs ( $n = 11$ ) of shop owners and/or outlet workers, slaughtering blade swab ( $n = 10$ ), and bleeding cone swab ( $n = 15$ ) were also collected in 3 mL of LB broth (Becton, Dickinson and Company, Franklin Lakes, NJ, USA) using sterile cotton buds, and were transported to the laboratory under cold condition. Likewise, swab samples collected in LB broth were incubated overnight (~20 h) at 37 °C for enrichment.

### 2.3. Detection and isolation of *E. albertii*

After enrichment, DNA was extracted from 100 µL of the culture by alkaline heat-extraction method. Briefly, bacterial cells were pelleted by centrifuging 100 µL of the enrichment culture at ca. 10,000 g for 3 min. The pellet was then resuspended in 85 µL of 50 mM NaOH, boiled for 10 min, and cooled on ice. Afterward 15 µL of 1 M Tris-HCl buffer (pH 7.8) was added, and the mixture was centrifuged at 16,000 g for 5 min at 4 °C. The resulting supernatant was collected and used as the DNA template for PCR. DNA template from enrichment cultures of the swabs was prepared by a boiling method (Bag et al., 2021). The DNA templates were subjected to an *E. albertii*-specific PCR assay targeting *Eacdt* genes for the detection of *E. albertii* (Hinenoya et al., 2019b).

When the *Eacdt* gene was detected by the PCR, the corresponding enrichment culture was spread onto XRM-MacConkey (XRM-MAC) agar plates for isolating *E. albertii* (Hinenoya et al., 2020b), and the plates were incubated at 37 °C overnight (~20 h). Potential *E. albertii* colorless colonies (2–12 colonies per sample) were screened for the *Eacdt* gene and the gene-positive colonies were streaked onto XRM-MAC plates for single isolation. Obtained colonies were confirmed as *E. albertii* by another PCR assay which can differentiate 3 *Escherichia* species including *E. albertii*, *E. coli* and *E. fergusonii* (Lindsey et al., 2017). The *E. albertii* isolates were cultured in 3 mL of LB-broth at 37 °C overnight (~20 h) and kept with glycerol (final conc. 20 %) at –80 °C until use.

## 2.4. Enterobacterial repetitive intergenic consensus (ERIC)-PCR

ERIC-PCR was applied for molecular typing of *E. albertii* isolates (Versalovic et al., 1991) with minor modifications. The reaction mixture had a final volume of 20 µL consisting of 10 µL GoTaq® Green 2× master mix (Promega Corp., Madison, WI, USA), 10 pmol of each primer, 1 µL of DNA template. The PCR was performed on a thermal cycler (Turbo-Cycler Lite; BLUE-RAY Biotech, Zhongshan, Taipei, Taiwan) with an initial denaturation of 95 °C for 2 min, followed by 35 cycles of denaturation at 98 °C for 10 s, annealing at 52 °C for 1 min, and extension at 72 °C for 2.5 min, with a final extension at 72 °C for 3 min.

## 2.5. Pulsed field gel electrophoresis (PFGE)

DNA fingerprint of the bacterium was analyzed by PFGE according to the protocol of pulse Net USA with slight modifications. Briefly, freshly cultured bacterial isolates were embedded into 0.5 % SeaKem Gold agarose (Lonza Rockland, Inc., Rockland, ME, USA) followed by in situ lysis in cell lysis buffer (50 mM Tris-HCl [pH 8.0], 50 mM EDTA) containing 0.5 mg/mL Proteinase K (P8044-5G, Sigma-Aldrich Co. LLC, St. Louis, MO, USA) and 1.0 % N-laurylsarcosine (Sigma-Aldrich) at 54 °C for 2 h. The agarose-embedded genomic DNA was digested with 20 U of *Xba*I (Takara Bio., Shiga, Japan) at 37 °C for 2 h. The digested DNA was electrophoresed on a 1.0 % pulsed-field certified agarose gel (Bio-Rad laboratories Inc., Hercules, CA, USA) in 0.5× TBE buffer under auto-algorithm with a molecular weight ranging from 30 to 600 kb using a CHEF Mapper (Bio-Rad laboratories Inc.) *Xba*I-digested *S. Braenderup* strain H9812 was used as a molecular size marker.

## 2.6. Antimicrobial susceptibility testing

Selected *E. albertii* isolates were subjected to antimicrobial susceptibility testing. The antimicrobial susceptibility of the isolated *E. albertii* was determined by the Kirby Bauer disc diffusion method according to the Clinical and Laboratory Standards Institute guidelines (M100-Ed30) (CLSI, 2020) and interpreted based on the diameter size of inhibition zone as susceptible, intermediate and resistant. A total of 15 antimicrobials including ampicillin (AMP, 10 µg), cefotaxime (CTX, 30 µg), ceftazidime (CAZ, 30 µg), cefoxitin (FOX, 30 µg), fosfomycin (FOF, 50 µg), meropenem (MEM, 10 µg), imipenem (IPM, 10 µg), gentamicin (GEN, 10 µg), kanamycin (KAN, 30 µg), streptomycin (STR, 10 µg), tetracycline (TET, 30 µg), chloramphenicol (CHL, 30 µg), nalidixic acid (NAL, 30 µg), ciprofloxacin (CIP, 5 µg), sulfamethoxazole-trimethoprim (SXT, 23.75/1.25 µg) were tested for each isolate. The antimicrobial discs were purchased from Becton, Dickinson and Company. *E. coli* strain ATCC25922 was used as the control strain in each experiment. Isolates showing resistance to three or more classes of antimicrobials are considered multidrug resistant (MDR) (Magiorakos et al., 2012).

## 2.7. Detection of virulence genes

*E. albertii* isolates were screened for the presence of *eae* and *stx* genes by a PCR assay which can specifically detect *eae*, *stx* (both *stx1* and *stx2*) and *cdt* genes (Hassan et al., 2019).

## 2.8. Whole genome sequencing (WGS) and analysis

Short-read sequencing was performed on NextSeq 1000 sequencing system. *E. albertii* isolates were cultured in 3 mL of LB broth (Becton, Dickinson and Company) and genomic DNA was extracted using DNeasy Blood & Tissue Kits following the protocols for Gram-negative bacteria (Qiagen, Hilden, Germany). DNA library was prepared by dual indexing using QIAseq FX DNA Library CDI Kit (QIAGEN). DNA concentration of the library was measured by Qubit fluorometer (Thermo Fisher Scientific, Waltham, MA, USA). A 150 bp pair-end sequencing was done on NextSeq 1000 sequencer using a NextSeq 1000/2000 P1 reagent

cartridge (300 cycles). Fastp v0.24.0 was used for trimming and filtering of raw read sequences (Chen, 2023), and SKESA v2.5.1 was used for de novo assembling with default parameters (Souvorov et al., 2018). The draft sequences were applied as the draft genomes to the following analyses. Virulence gene profiling was done by uploading the draft genomes to Virulence Factor Database (VFDB; <http://www.mgc.ac.cn/VF/>) using the Genus *Escherichia* database (Liu et al., 2022a). AMR gene, plasmid profiling and cgMLST were done on the online platform of Center for Genomic Epidemiology (<https://www.genomicepidemiology.org/services/>) using the ResFinder v4.6.0 (Bortolaia et al., 2020; Camacho et al., 2009), PlasmidFinder v2.1 with the Enterobacteriales database (Camacho et al., 2009; Carattoli et al., 2014), and cgMLSTfinder v1.0.1 (Clausen et al., 2018; Jolley and Maiden, 2010), respectively. EAO and EAH genotypes were determined by in silico analysis. Sequencing reads of each isolate were mapped to *wzx/wzy* gene sequences of the 40 EAOs and *fliC* gene sequences of the 4 EAHs from each reference strain, respectively (Nakae et al., 2021; Ooka et al., 2019) on CLC Genomics Workbench 24 (Qiagen). For phylogenetic analysis, Snippy v3.1 was used to identify single nucleotide polymorphisms (SNPs) in the genomes of the *E. albertii* strains and generate the SNP alignment using the genome of *E. albertii* strain CB9786 (Accession number AP014856.1) as reference. Prophage regions were identified using PHASTEST (<https://phastest.ca/>) and masked from the alignment. Recombinant regions were identified and removed using Gubbins v3.4 (Croucher et al., 2014). The final core alignment of 45,375 bases was extracted with SNP-sites (Page et al., 2016). A maximum likelihood phylogenetic tree was inferred under the best-fit model of TVM + G with 1000 bootstrap replicates using RAXML Next Generation v1.2.2 (Kozlov et al., 2019). For pairwise SNP distance calculation, core genome alignment was performed through Roary Pangenome pipeline (Page et al., 2015), and SNP distance was calculated from the core genome alignment using snp-dists (<https://github.com/tseemann/snp-dists>) and visualized with plot\_matrix\_heatmap.sh ([https://ead-csic-compbio.github.io/get\\_homologues/manual/](https://ead-csic-compbio.github.io/get_homologues/manual/)). The mid-point rooted tree was visualized using iTOL ver. 6 (Letunic and Bork, 2024) together with other characteristics and sample information of each isolate.

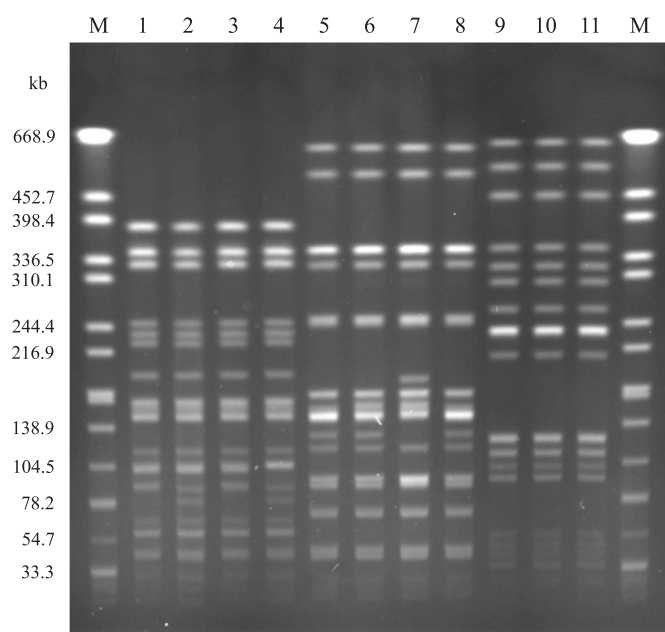
## 3. Results

### 3.1. Occurrence of *E. albertii* in retail chickens, handlers and equipment

To know the distribution of *E. albertii* in retail chicken meats in Bangladesh, initially chicken meats were purchased by the whole body 2 times at 2 outlets (A and B) in K. R. market of Sadar upazila, and 4 parts of meats including breast, thigh, P-G and liver from each chicken were individually tested for the presence of *E. albertii* after enrichment (Table 1). Although *E. albertii* was not detected in the meats from all the 4 chickens of outlet A by an *E. albertii*-specific PCR assay, those from the chickens ( $n = 5$ ) of outlet B were constantly 100 % PCR-positive, suggesting that occurrence of *E. albertii* could be different by outlets. Then, chicken samples were extensively collected from outlets (H to Q; 15 from 4 upazila) in different locations. From November 2022, swab was also collected from cloaca of chickens, hands of handlers, and blades and exsanguination cones for slaughtering, respectively. While no meats were PCR-positive at 3 outlets (H, J and Q), any of the meats were PCR-positive in all the tested chickens at the remaining 12 outlets. Consistent with the *E. albertii* detection in the meats, corresponding cloacal swabs were also positive by the PCR assay except for the chickens of outlet Q, suggesting the contamination of meats by intestinal contents containing *E. albertii* during processing. Accordingly, when the meats were PCR-positive, any of the surrounding environmental samples including hands, cones and blades were also positive for *E. albertii* detection. *E. albertii* was detected in the meats of 60.7 % thigh, 59.0 % breast, 57.4 % liver and 50.8 % P-G (Table 1), where no statistical difference was found in the occurrence of *E. albertii* among the parts ( $p = 0.804$  by Fisher's exact test). Using an *E. albertii*-selective isolation medium, XRM-







**Fig. 2.** Genetic relatedness of the *Escherichia albertii* isolates from chickens and handlers.

Genomic DNA was digested with *Xba*I and separated by PFGE. Lanes 1, Ha5-1 (hand); 2, 39Clo-1 (cloaca); 3, 39B-1 (breast); 4, 40B-1 (breast); 5, Ha-6 (hand); 6, 42Clo-1 (cloaca); 7, 42B-1 (breast); 8, 33Clo (cloaca); 9, Ha-7 (hand); 10, 48Clo (cloaca); 11, 48T-1 (thigh). M, *Xba*I-digested genomic DNA of *Salmonella* Braenderup strain H9812 used as a molecular size marker.

1), the isolates from their cloacae and meats were clonally related with 0 to 3 pairwise SNP distances (Supplementary Fig. 3). The handler's isolate Ha-6 was also clonally related to the isolates from chicken ID 42 that the handler processed although the handler's isolate Ha-5 showed 33 to 40 SNP distances with the isolates from the chickens that the handler processed. Furthermore, clonal relationships were found between the meat isolates from different chicken individuals in same sampling batch (17B-7 vs 18T-1 vs 21L-8 from outlet F, and 3T-1 vs 4L-1 vs 4B-1 from outlet B) although cloacal and handler's samplings were not done at that time. Interestingly, the handler's isolate Ha-7 from outlet M was also clustered together with the chicken isolates from another outlet N (48Clo and 48T-1). These data indicate a possible cross contamination among handlers and chickens. The genomic similarity of the isolates was further confirmed by PFGE (Fig. 2). Isolates from the chicken meat, cloacal and human hand swab revealed identical PFGE patterns. Some of them showed one band difference each other, but they can be interpreted as clonal considering the Tenover's criteria (Tenover et al., 1995). Moreover, genetic relatedness was also observed between isolates from different Market (33Clo vs Ha-6 vs 42Clo vs 42B).

Analysis of AMR genes revealed the presence of genes against respective AMR-phenotypes (Fig. 1, Supplementary Table 1), while the isolates susceptible to all the antimicrobials tested didn't carry any known AMR genes. Among the 34 TET resistant isolates, 88 % carried *tetA* and the rest (12 %) carried *tetB* gene. On the other hand, quinolone resistant isolates showed different combinations of chromosomal gene mutations as well as plasmid-mediated quinolone resistance (PMQR) genes. All the 25 quinolone resistant isolates carried mutation in at least one of chromosomal resistance genes (*gyrA*, *parE*, *parC*), and/or harbored PMQRGs (*qnrS*, *qepA*). S83L mutation in *gyrA* gene, *gyrA* (S83L) was found in all the 25 isolates, but not in the isolates with no resistance to quinolones. *parE* (I355L) mutation was found in 7 quinolone-resistant isolates while the mutation was also found in 6 quinolone-sensitive isolates. Two quinolone resistant isolates additionally harbored *parC* (S80I) mutation. *qnrS* (*qnrS1* or *qnrS13*) gene was found in 11 resistant and 1 susceptible isolate to quinolones,

respectively. *qepA3* gene was also detected in 2 quinolone resistant isolates.

All the ampicillin resistant isolates carried *bla*<sub>TEM-1B</sub>, and the SXT resistant isolates carried *sul2* and *dfrA* (*dfrA1* or *dfrA14*) genes, indicating their association with the respective antimicrobial resistances. Besides, aminoglycosides resistance genes *aph*(4)*Ia*, *aph*(6)*-Id*, *aph*(3')*-Ia*, *aph*(3')*-Ib* and *aac*(3)-IV were detected in 2 isolates showing resistance to respective antimicrobials (Fig. 1, Supplementary Table 1). The 2 isolates additionally carried hygromycin-resistance gene, *aph*(4)*-Ia*, and macrolide-resistance gene, *mph*(A) gene although the phenotypes was not tested.

Plasmid typing of the isolates was further performed using a Plasmid Finder software, and various plasmid replicons were identified in the isolates, such as Col156, IncFIA, IncFIB, IncFII, IncX1, IncX4, IncI1, IncHI2, IncHI2A and IncX1 (Supplementary Table 1). Interestingly, *qnrS* and *tetA* genes were located on the same contigs with the replicon IncI1 in all the nine isolates of cgST 58,443 and an isolate of cgST46866. The three AMR genes (*bla*<sub>TEM-1B</sub>, *qnrS1* and *tetA*) of isolate 17B-3 shared the contig with the replicon IncX1. Furthermore, six AMR genes (*bla*<sub>TEM-1B</sub>, *strA*, *strB*, *qnrS1*, *sul2* and *tetA*) shared the contig with the replicon IncFII in the isolate 47Clo-1.

#### 4. Discussion

This study highlights the significant occurrence of *E. albertii* in retail chicken outlets in Bangladesh, with 63.9 % of chickens testing positive in their meats. The detection of *E. albertii* in multiple anatomical sites, including thigh muscle, breast, liver, and P-G, suggests widespread contamination across the meat products examined. Notably, cross-contamination was evident, as isolates from chicken meat, cloacal swabs, and human hand swabs from the same retail outlets were genetically identical, underscoring the importance of hygienic practices during meat processing and handling (Ferdous et al., 2023). To the best of our knowledge, this is the first report of *E. albertii* from retail chicken meats in Bangladesh, which raises concerns about the potential risks to public health.

The prevalence of *E. albertii* in our study is considerably higher than that reported in similar studies globally. For instance, previous studies have identified *E. albertii* as an emerging pathogen in wild animals, birds, poultry, poultry carcass washings and occasionally in humans, but its detection in retail meat, especially at such high rates, is uncommon (Hinenoya et al., 2021; Hinenoya et al., 2020a; Hinenoya et al., 2022b; Lindsey et al., 2015; Maeda et al., 2015; Ooka et al., 2013). The occurrence of *E. albertii* in chicken meats and giblets was reported as 2.83–2.88 % in Japan (Konno et al., 2021; Maeda et al., 2015). In addition to meats, this study detected 71.4 % *E. albertii* from cloacal swab which is much higher than those reported in cloacal swab of broiler chicken in the USA (15.92 to 30 %) (Hinenoya et al., 2021; Wang et al., 2022), and backyard poultry (3.5 %) in Bangladesh (Gupta et al., 2023). The high level of difference might be associated with the sample size or geographical location, however, association of poor hygiene and production practices at pre-harvest level cannot be overlooked. In addition, we encountered a significant difference in the *E. albertii* occurrence at different locations. The significant differences in occurrence between sampling locations also suggest potential variations in local handling practices or environmental factors that may influence the spread of this pathogen. These findings are a crucial addition to the growing body of literature on the role of foodborne pathogens in disease transmission.

A concerning finding from our study is the high level of MDR exhibited by *E. albertii* isolates, with 50.0 % of the strains being classified as MDR. Resistance to antimicrobials such as TET, NAL, AMP, CIP and GEN is alarming, particularly in the context of Bangladesh, where use of these antimicrobials is widespread and officially approved for use in animal production (DGDA, 2023; Foysal et al., 2024; Hoque et al., 2020;

**Table 1**  
Occurrence of *Escherichia albertii* in broiler chickens and the retail outlets in Mymensingh, Bangladesh.

| Location   | Retail markets    | Outlet   | Collection date (YYYY/MM/DD) | No. of chickens |                                       | No. of PCR-positive samples/No. of <i>E. albertii</i> isolation-positive samples |             |             |             |             | No. of PCR-positive/No. of samples (%) /No. of <i>E. albertii</i> isolation-positive |          |          |
|------------|-------------------|----------|------------------------------|-----------------|---------------------------------------|----------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|--------------------------------------------------------------------------------------|----------|----------|
|            |                   |          |                              | Tested          | <i>E. albertii</i> -positive in meats | Breast                                                                           | Thigh       | P-G         | Liver       | Cloaca      | Hand                                                                                 | Cone     | Blade    |
| Sadar      | K. R. market      | A        | 2021/10/06                   | 2               | 0 (0)                                 | 0/NA                                                                             | 0/NA        | 0/NA        | 0/NA        | ND          | ND                                                                                   | ND       | ND       |
|            |                   |          | 2022/01/01                   | 2               | 0 (0)                                 | 0/NA                                                                             | 0/NA        | 0/NA        | 0/NA        | ND          | ND                                                                                   | ND       | ND       |
|            |                   | B        | 2021/11/08                   | 2               | 2 (100)                               | 2/1                                                                              | 2/2         | 2/1         | 2/1         | ND          | ND                                                                                   | ND       | ND       |
|            |                   |          | 2022/01/07                   | 3               | 3 (100)                               | 3/0                                                                              | 3/0         | 3/0         | 3/0         | ND          | ND                                                                                   | ND       | ND       |
|            |                   | Subtotal |                              | 10              | 5 (50.0)                              | 5/1                                                                              | 5/2         | 5/1         | 5/1         | ND          | ND                                                                                   | ND       | ND       |
|            | Kewatkhali market | C        | 2022/01/21                   | 3               | 2 (66.7)                              | 1/0                                                                              | 1/0         | 1/0         | 0/NA        | ND          | ND                                                                                   | ND       | ND       |
|            |                   | D        | 2022/02/11                   | 3               | 3 (100)                               | 2/0                                                                              | 3/0         | 2/0         | 3/0         | ND          | ND                                                                                   | ND       | ND       |
|            |                   | E        | 2022/09/22                   | 3               | 1 (33.3)                              | 0/NA                                                                             | 1/0         | 0/NA        | 0/NA        | ND          | ND                                                                                   | ND       | ND       |
|            |                   | Subtotal |                              | 9               | 6 (66.7)                              | 3/0                                                                              | 5/0         | 3/0         | 3/0         | ND          | ND                                                                                   | ND       | ND       |
|            | Mechua Bazar      | F        | 2022/06/18                   | 5               | 5 (100)                               | 5/1                                                                              | 5/2         | 5/0         | 5/2         | ND          | ND                                                                                   | ND       | ND       |
|            |                   | G        | 2022/09/22                   | 2               | 2 (100)                               | 2/0                                                                              | 2/0         | 2/0         | 2/2         | ND          | ND                                                                                   | ND       | ND       |
|            |                   | H        | 2022/11/20                   | 5               | 0 (0)/NA                              | 0/NA                                                                             | 0/NA        | 0/NA        | 0/NA        | 0/NA        | 0/2                                                                                  | 0/3      | 0/1      |
|            |                   | Subtotal |                              | 12              | 7 (58.3)                              | 7/1                                                                              | 7/2         | 7/0         | 7/4         | 0/NA        | (0)/NA                                                                               | (0)/NA   | (0)/NA   |
| Tarakanda  | Tarakanda Bazar   | I        | 2022/12/06                   | 3               | 3 (100)                               | 3/2                                                                              | 3/2         | 3/1         | 3/1         | 3/3         | 0/1                                                                                  | 0/3      | 0/1      |
|            |                   | J        |                              | 4               | 0 (0)                                 | 0/NA                                                                             | 0/NA        | 0/NA        | 0/NA        | 0/NA        | (0)/NA                                                                               | (0)/NA   | (0)/NA   |
|            |                   | K        |                              | 3               | 3 (100)                               | 3/2                                                                              | 3/1         | 3/0         | 3/1         | 3/1         | 1/1                                                                                  | 1/1      | 0/1      |
|            |                   | Subtotal |                              | 10              | 6 (60.0)                              | 6/4                                                                              | 6/3         | 6/1         | 6/2         | 6/4         | 1/3                                                                                  | 1/6      | 0/3      |
|            |                   |          |                              |                 |                                       |                                                                                  |             |             |             |             | (33.3)/1                                                                             | (16.7)/0 | (0)/NA   |
| Trishal    | Trishal Bazar     | L        | 2023/01/08                   | 3               | 3 (100)                               | 3/1                                                                              | 3/0         | 3/0         | 3/0         | 3/1         | 1/1                                                                                  | 0/1      | 0/1      |
|            |                   | M        |                              | 3               | 3 (100)                               | 3/0                                                                              | 3/0         | 3/0         | 3/0         | 3/1         | (100)/1                                                                              | (0)/NA   | (0)/NA   |
|            |                   | N        |                              | 4               | 4 (100)                               | 4/0                                                                              | 4/1         | 4/0         | 4/0         | 3/1         | 1/1                                                                                  | 1/1      | 1/1      |
|            |                   | Subtotal |                              | 10              | 10 (100)                              | 10/1                                                                             | 10/1        | 10/0        | 10/0        | 9/3         | (100)/1                                                                              | (100)/1  | (100)/1  |
|            |                   |          |                              |                 |                                       |                                                                                  |             |             |             |             | 2/3                                                                                  | 1/3      | 1/3      |
| Muktagacha | Muktagachha Bazar | O        | 2023/03/02                   | 3               | 3 (100)                               | 3/ND                                                                             | 2/ND        | 0/ND        | 2/ND        | 3/ND        | (66.7)/2                                                                             | (33.3)/1 | (33.3)/1 |
|            |                   | P        |                              | 3               | 2 (66.7)                              | 2/ND                                                                             | 2/ND        | 0/NA        | 2/ND        | 3/ND        | 1/1                                                                                  | 0/1      | 0/1      |
|            |                   | Q        |                              | 4               | 0 (0)                                 | 0/NA                                                                             | 0/NA        | 0/NA        | 0/NA        | 4/ND        | (100)/ND                                                                             | (0)/NA   | (0)/NA   |
|            |                   | Subtotal |                              | 10              | 5 (50)                                | 5/ND                                                                             | 4/ND        | 0/ND        | 4/ND        | 10/ND       | 2/3                                                                                  | 0/3      | 0/3      |
|            |                   |          |                              |                 |                                       |                                                                                  |             |             |             |             | (66.7)/ND                                                                            | (0)/NA   | (0)/NA   |
| Total      |                   |          |                              | 61              | 39 (63.9)                             | 36 (59.0)/7                                                                      | 37 (60.7)/8 | 31 (50.8)/2 | 35 (57.4)/7 | 25 (71.4)/7 | 5/11                                                                                 | 2/15     | 1/10     |
|            |                   |          |                              |                 |                                       |                                                                                  |             |             |             |             | (45.5)/3                                                                             | (13.3)/1 | (10.0)/1 |

ND, not done; NA, not applicable (i.e. the samples were negative by *E. albertii*-specific PCR thus not subjected for the isolation of *E. albertii*); P-G, proventriculus gizzard.

Imam et al., 2023). According to Bangladesh National Veterinary Formulary (BDNVF) (DGDA, 2023), several antimicrobials have been routinely used for the growth and production as well as treatments of animals including livestock, in which all the antimicrobials to which the *E. albertii* isolates in this study frequently showed resistances are included. Such resistance patterns are consistent with previous studies on other foodborne pathogens in Bangladesh, reinforcing concerns about the public health implications of antimicrobial use in poultry farming (Ali et al., 2023; Foysal et al., 2024; Imam et al., 2023). The detection of AMR genes further emphasizes the risk of antimicrobial-resistant *E. albertii* being transmitted through the food chain. Interestingly, we observed multiple avenues of achieving quinolone resistance in the study isolates. The isolates carried PMQR genes alongside chromosomal mutations in *gyrA*, *parE* and *parC*. Although

PMQR genes have been identified, such as *qnr*, *qep* and *aac(6)-Ib-cr* (Jacoby et al., 2014), major mechanism of quinolone resistance is chromosomal gene mutations encoding for DNA gyrase subunits (*gyrA* and *gyrB*) and topoisomerase IV subunits (*parC* and *parE*), particularly in the quinolone resistance determining regions (QRDRs). Various mutations associated with resistance to quinolones have been identified (Boueroy et al., 2023; Sáenz et al., 2003), and accumulation of the mutations makes the resistance level of bacteria higher (Bagel et al., 1999; Huseby et al., 2017). In this study, all the whole genome-sequenced isolates harbored mutations in one or more of the genes associated with quinolone resistance, suggesting that the routine use of antimicrobials has enhanced the selection of the strains with resistance to quinolones. Indeed, 70.3 % and 21.6 % of the *E. albertii* isolates were resistant to NAL and CIP, respectively. While sole mutation of *parE*

**Table 2**  
Antimicrobial resistance pattern of the *Escherichia albertii* isolates.

| Resistance pattern                     | No. of antimicrobial class | No. of isolates |
|----------------------------------------|----------------------------|-----------------|
| TET, GEN, KAN, STR, AMP, CIP, NAL, SXT | 5                          | 2               |
| TET, AMP, CIP, NAL, SXT                | 4                          | 4               |
| TET, STR, AMP, SXT                     | 4                          | 1               |
| TET, AMP, SXT                          | 3                          | 6               |
| TET, GEN, NAL                          | 3                          | 5               |
| TET, AMP                               | 2                          | 1               |
| TET, NAL                               | 2                          | 14              |
| TET                                    | 1                          | 1               |
| None                                   | 0                          | 2               |
| Total                                  |                            | 36              |

(I355T) was identified in the isolates with and without resistance to NAL, all the NAL-resistant isolates harbored *gyrA* (S83L) gene mutation. Considering the mutation is the major quinolone resistance mechanism of *E. coli* (Boueroy et al., 2023; Huseby et al., 2017; Sáenz et al., 2003), the *gyrA* (S83L) gene mutation could also be responsible for the NAL resistance of the *E. albertii* isolates in this study. The *E. albertii* isolates with resistance to CIP showed additional common genetic features of harboring a mutation of *parE* (I355T) and carrying *qnrS* gene, suggesting that either or both the genes decreased the susceptibility to quinolones, which lead to CIP resistance of the isolates. To the best of our knowledge, this is the first report showing the mutations in both *gyrA* and *parE* gene mutations in *E. albertii*. However, since this study did not quantitatively analyze the quinolone resistances, it cannot be explained how much they are associated with the quinolone resistance of the *E. albertii* isolates. Thus, further studies are needed to unveil their contribution in resistance by gene-manipulation works. Moreover, although multiple mutations decreasing the susceptibility to quinolones are frequently identified in the chromosomal genes for quinolone resistance in *E. coli* (Boueroy et al., 2023; Sáenz et al., 2003), only single mutation has been identified in those genes of *E. albertii* including the isolates of the present study (Iguchi et al., 2023). However, the information is severely limited in *E. albertii*. Continuous and comprehensive epidemiological investigations are highly warranted to understand the real situation of the quinolone resistances in this bacterium.

In addition to quinolones resistance genes, *tetA*, *tetB*, *bla*<sub>TEM-1B</sub>, and *sul2* were detected in the *E. albertii* isolates. The association of *qnrS* and *tetA* genes with IncFII and IncI1 plasmids suggests that these mobile genetic elements play a key role in the horizontal transfer of resistance genes, further emphasizing the risk of transferring these resistances through food chain. Such resistances and associated genotypes were also reported in *E. albertii* isolated from human and animals including chicken (Bengtsson et al., 2023; Li et al., 2018). Tetracycline, beta-lactams and sulfonamides are among the mostly used antimicrobials in the poultry industries of Bangladesh, and occurrence of corresponding resistance genes have been reported from food borne pathogens isolated from cloacal swab, meat and chicken rearing environment in Bangladesh (Ali et al., 2023; Foysal et al., 2024; Parvin et al., 2022), indicating a widespread distribution of these resistance genes in the environment and food borne pathogens. In addition, the presence of plasmid types such as IncX1 and IncX4, which have been frequently linked to the dissemination of AMR genes in other pathogens (Dobiasova and Dolejska, 2016; Juraschek et al., 2021) further highlights the potential for the spread of AMR through the food supply chain.

The genomic characterization of *E. albertii* isolates revealed genetic diversity, as evidenced by the clustering patterns based on cgMLST and EAO:H genotyping. Despite this diversity, isolates from the same retail outlets, and even the same chicken were genetically identical, indicating potential cross-contamination during processing. In the present study, the genetic similarity among same EAOg:Hg type was further confirmed by PFGE analysis, which demonstrated identical PFGE patterns among isolates from chicken meat, cloacal swabs, and human hand swabs.

These findings strongly suggest that poor hygienic practices during handling and processing may facilitate the transmission of *E. albertii* from contaminated chicken to human hands, posing a risk of human infections.

The findings of this study have important public health implications, particularly in Bangladesh, where poultry is a major component of the diet and foodborne pathogens remain a significant concern. The high prevalence of *E. albertii* in retail chicken meat, combined with the evidence of cross-contamination during processing, raises serious concerns about food safety. The transmission of MDR *E. albertii* from poultry to humans, as suggested by the identical genomic and phenotypic characteristics of isolates from chicken and human hand swabs, underscores the need for improved hygiene and biosecurity measures in retail meat outlets. Public health interventions, including education on proper handling practices and routine monitoring of foodborne pathogens, are essential to mitigate the risks associated with contaminated poultry products.

## 5. Conclusions

This study provides compelling evidence of the widespread occurrence of *E. albertii* in retail chicken meat in Bangladesh and the potential for cross-contamination during processing. The high rates of multidrug resistance, coupled with the presence of virulence genes, highlight the need for stringent food safety measures and continued surveillance of emerging pathogens like *E. albertii*. Future research should focus on the sources of contamination at the farm level, as well as the potential human health risks associated with the consumption of contaminated poultry products.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijfoodmicro.2025.111081>.

## CRedit authorship contribution statement

**Jayedul Hassan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kishor Sasmith Utsho:** Investigation, Formal analysis, Data curation. **Susmita Karmakar:** Investigation, Data curation. **Md. Wohab Ali:** Investigation, Data curation. **Sharda Prasad Awasthi:** Investigation, Formal analysis, Data curation. **Chiharu Uyama:** Investigation, Formal analysis. **Noritoshi Hatanaka:** Data curation. **Shinji Yamasaki:** Writing – review & editing, Formal analysis, Data curation. **Atsushi Hinenoya:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

## Funding information

This work was supported by basic research grant from Osaka Metropolitan University, the Japan Society for the Promotion of Science KAKENHI (grant numbers 20K06396, 24K09249) and a grant from Ohyama Health Foundation Inc. to AH, and the Ministry of Education (MoE), Government of the People's Republic of Bangladesh, through the Grants for Advanced Research in Education (GARE) (Project no. LS20191223) to JH.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



## Data availability

All sequence data obtained in this study have been deposited in DDBJ under BioProject accession number PRJDB19619.

## References

- Albert, M.J., Alam, K., Islam, M., Montanaro, J., Rahaman, A.S., Haider, K., Hossain, M. A., Kibriya, A.K., Tzipori, S., 1991. *Hafnia alvei*, a probable cause of diarrhea in humans. *Infect. Immun.* 59, 1507–1513. <https://doi.org/10.1128/iai.59.4.1507-1513.1991>.
- Ali, M.W., Utsho, K.S., Karmakar, S., Hoque, M.N., Rahman, M.T., Hassan, J., 2023. First report on the molecular characteristics of *mcr-1* colistin resistant *E. coli* isolated from retail broiler meat in Bangladesh. *Int. J. Food Microbiol.* 388, 110065. <https://doi.org/10.1016/j.ijfoodmicro.2022.110065>.
- Asoshima, N., Matsuda, M., Shigemura, K., Honda, M., Yoshida, H., Oda, T., Hiwaki, H., 2015. Isolation of *Escherichia albertii* from raw chicken liver in Fukuoka city, Japan. *Jpn. J. Infect. Dis.* 68, 248–250. <https://doi.org/10.7883/yoken.JJID.2014.530>.
- Bag, M.A.S., Khan, M.S.R., Sami, M.D.H., Begum, F., Islam, M.S., Rahman, M.M., Rahman, M.T., Hassan, J., 2021. Virulence determinants and antimicrobial resistance of *E. coli* isolated from bovine clinical mastitis in some selected dairy farms of Bangladesh. *Saudi. J. Biol. Sci.* 28, 6317–6323. <https://doi.org/10.1016/j.sjbs.2021.06.099>.
- Bagel, S., Hüllen, V., Wiedemann, B., Heisig, P., 1999. Impact of *gyrA* and *parC* mutations on quinolone resistance, doubling time, and supercoiling degree of *Escherichia coli*. *Antimicrob. Agents Chemother.* 43, 868–875. <https://doi.org/10.1128/aac.43.4.868>.
- Barmettler, K., Biggel, M., Treier, A., Muchaamba, F., Vogler, B.R., Stephan, R., 2022. Occurrence and characteristics of *Escherichia albertii* in wild birds and poultry flocks in Switzerland. *Microorganisms* 10, 2265. <https://doi.org/10.3390/microorganisms10112265>.
- Barmettler, K., Biggel, M., Treier, A., Muchaamba, F., Stephan, R., 2023. Livestock as possible reservoir of *Escherichia albertii* in Switzerland. *Schweiz. Arch. Tierheilkd.* 165, 299–306. <https://doi.org/10.17236/sat00393>.
- Bengtsson, R.J., Baker, K.S., Cunningham, A.A., Greig, D.R., John, S.K., Macgregor, S.K., Seilern-Moy, K., Spiro, S., Chong, C.C., De Silva, P.M., Jenkins, C., Lawson, B., 2023. The genomic epidemiology of *Escherichia albertii* infecting humans and birds in Great Britain. *Nat. Commun.* 14, 1707. <https://doi.org/10.1038/s41467-023-37312-3>.
- Bhatt, S., Egan, M., Critelli, B., Kouse, A., Kalman, D., Upreti, C., 2019. The evasive enemy: insights into the virulence and epidemiology of the emerging attaching and effacing pathogen *Escherichia albertii*. *Infect. Immun.* 87. <https://doi.org/10.1128/IAI.00254-18.e00254-00218>.
- Bortolaia, V., Kaas, R.S., Ruppe, E., Roberts, M.C., Schwarz, S., Cattoir, V., Philippon, A., Allesoe, R.L., Rebelo, A.R., Florensa, A.F., Fagelhauer, L., Chakraborty, T., Neumann, B., Werner, G., Bender, J.K., Stingl, K., Nguyen, M., Coppens, J., Xavier, B. B., Malhotra-Kumar, S., Westh, H., Pinholt, M., Anjum, M.F., Duggett, N.A., Kempf, I., Nykäsenoja, S., Olkkola, S., Wiecek, K., Amaro, A., Clemente, L., Mossong, J., Losch, S., Ragimbeau, C., Lund, O., Aarestrup, F.M., 2020. ResFinder 4.0 for predictions of phenotypes from genotypes. *J. Antimicrob. Chemother.* 75, 3491–3500. <https://doi.org/10.1093/jac/dkaa345>.
- Bouero, P., Chopjitt, P., Hatrongjit, R., Morita, M., Sugawara, Y., Akeida, Y., Iida, T., Hamada, S., Kerdin, A., 2023. Fluoroquinolone resistance determinants in carbapenem-resistant *Escherichia coli* isolated from urine clinical samples in Thailand. *PeerJ* 11, e16401. <https://doi.org/10.7717/peerj.16401>.
- Brandal, L.T., Tunjso, H.S., Ranheim, T.E., Loberli, I., Lange, H., Wester, A.L., 2015. Shiga toxin 2a in *Escherichia albertii*. *J. Clin. Microbiol.* 53, 1454–1455. <https://doi.org/10.1128/JCM.03378-14>.
- Camacho, C., Coulouris, G., Avagyan, V., Ma, N., Papadopoulos, J., Bealer, K., Madden, T.L., 2009. BLAST+: architecture and applications. *BMC Bioinformatics* 10, 421. <https://doi.org/10.1186/1471-2105-10-421>.
- Carattoli, A., Zankari, E., García-Fernández, A.G.-F., Voldby Larsen, M., Lund, O., Villa, L., Møller Aarestrup, F., Hasman, H., 2014. In silico detection and typing of plasmids using PlasmidFinder and plasmid multilocus sequence typing. *Antimicrob. Agents Chemother.* 58, 3895–3903. <https://doi.org/10.1128/aac.02412-14>.
- Chen, S., 2023. Ultrafast one-pass FASTQ data preprocessing, quality control, and deduplication using fastp. *Imeta* 2, e107. <https://doi.org/10.1002/imt2.107>.
- Chowdhury, S., Azziz-Baumgartner, E., Kile, J.C., Hoque, M.A., Rahman, M.Z., Hossain, M.E., Ghosh, P.K., Ahmed, S.S.U., Kennedy, E.D., Sturm-Ramirez, K., Gurley, E.S., 2020. Association of biosecurity and hygiene practices with environmental contamination with Influenza A viruses in live bird markets, Bangladesh. *Emerg. Infect. Dis.* 26, 2087–2096. <https://doi.org/10.3201/eid2609.191029>.
- Clausen, P.T.L.C., Aarestrup, F.M., Lund, O., 2018. Rapid and precise alignment of raw reads against redundant databases with KMA. *BMC Bioinformatics* 19, 307. <https://doi.org/10.1186/s12859-018-2336-6>.
- CLSI, 2020. Performance Standards for Antimicrobial Susceptibility Testing, 30th edition M100-Ed30. Clinical And Laboratory Standards Institute. <https://webstore.ansi.org/standards/cls/clsim100ed30>.
- Croucher, N.J., Page, A.J., Connor, T.R., Delaney, A.J., Keane, J.A., Bentley, S.D., Parkhill, J., Harris, S.R., 2014. Rapid phylogenetic analysis of large samples of recombinant bacterial whole genome sequences using Gubbins. *Nucleic Acids Res.* 43, e15. <https://doi.org/10.1093/nar/gku1196>.
- DGDA, 2023. Bangladesh National Veterinary Formulary 2023. Directorate General of Drug Administration, Ministry of Health & Family Welfare, Bangladesh. <https://file-mymensingh.portal.gov.bd/uploads/2696f934-1fec-4d0b-ac6d-772cbe52613a//663/9d2/cc5/6639d2cc5d66c66549883.pdf>.
- Dobiasova, H., Dolejska, M., 2016. Prevalence and diversity of IncX plasmids carrying fluoroquinolone and  $\beta$ -lactam resistance genes in *Escherichia coli* originating from diverse sources and geographical areas. *J. Antimicrob. Chemother.* 71, 2118–2124. <https://doi.org/10.1093/jac/dkw144>.
- Ferdous, J., Uddin, M.H., Mahmud, R., Hennessey, M., Al Sattar, A., Das Gupta, S., Gibson, J.S., Alders, R., Henning, J., Fournié, G., Hoque, M.A., 2023. Mapping of dressed and processed poultry products in Bangladesh: identifying the food safety risks for policy intervention. *Vet. Res. Commun.* 47, 1991–2002. <https://doi.org/10.1007/s11259-023-10153-7>.
- Foyals, M., Imam, T., Das, S.B., Gibson, J.S., Mahmud, R., Gupta, S.D., Fournié, G., Hoque, M.A., Henning, J., 2024. Association between antimicrobial usage and resistance on commercial broiler and layer farms in Bangladesh. *Front. Vet. Sci.* 11, 1435111. <https://doi.org/10.3389/fvets.2024.1435111>.
- Gordon, D.M., 2011. Reservoirs of Infection: The Epidemiological Characteristics of an Emerging Pathogen, *Escherichia albertii*. Department of Agriculture, Fisheries and Forestry. Division of Ecology, Evolution and Genetics; Research School of Biology; The Australian National University Canberra.
- Gupta, M.D., Shaha, M., Lima, A., Ghosh, K., Logno, T.A., Das, A., 2023. Occurrence and antibiogram of *Escherichia albertii* in backyard poultry and pigeons in Bangladesh. *Int. J. One Health.* 9, 115–121. <https://doi.org/10.14202/IJOH.2023.115-121>.
- Hassan, J., Awasthi, S.P., Hatanaka, N., Okuno, K., Hoang, P.H., Nagita, A., Hinenoya, A., Yamasaki, S., 2019. Development of a multiplex PCR targeting *eae*, *stx* and *cdt* genes in genus *Escherichia* and detection of a novel *cdtB* gene in *Providencia rustigianii*. *Pathog. Dis.* 76. <https://doi.org/10.1093/femspd/ftz002>.
- Hinenoya, A., Yasuda, N., Hibino, T., Shima, A., Nagita, A., Tsukamoto, T., Yamasaki, S., 2017a. Isolation and characterization of an *Escherichia albertii* strain producing three different toxins from a child with diarrhea. *Jpn. J. Infect. Dis.* 70, 252–257. <https://doi.org/10.7883/yoken.JJID.2016.186>.
- Hinenoya, A., Yasuda, N., Mukaizawa, N., Sheikh, S., Niwa, Y., Awasthi, S.P., Asakura, M., Tsukamoto, T., Nagita, A., Albert, M.J., Yamasaki, S., 2017b. Association of cytolethal distending toxin-II gene-positive *Escherichia coli* with *Escherichia albertii*, an emerging enteropathogen. *Int. J. Med. Microbiol.* 307, 564–571. <https://doi.org/10.1016/j.ijmm.2017.08.008>.
- Hinenoya, A., Ichimura, H., Awasthi, S.P., Yasuda, N., Yatsuyanagi, J., Yamasaki, S., 2019a. Phenotypic and molecular characterization of *Escherichia albertii*: further surrogates to avoid potential laboratory misidentification. *Int. J. Med. Microbiol.* 309, 108–115. <https://doi.org/10.1016/j.ijmm.2018.12.003>.
- Hinenoya, A., Ichimura, H., Yasuda, N., Harada, S., Yamada, K., Suzuki, M., Iijima, Y., Nagita, A., Albert, M.J., Hatanaka, N., Awasthi, S.P., Yamasaki, S., 2019b. Development of a specific cytolethal distending toxin (*cdt*) gene (*Eacdt*)-based PCR assay for the detection of *Escherichia albertii*. *Diagn. Microbiol. Infect. Dis.* 95, 119–124. <https://doi.org/10.1016/j.diagmicrobio.2019.04.018>.
- Hinenoya, A., Nagano, K., Awasthi, S.P., Hatanaka, N., Yamasaki, S., 2020a. Prevalence of *Escherichia albertii* in raccoons (*Procyon lotor*). *Japan. Emerg. Infect. Dis.* 26, 1304–1307. <https://doi.org/10.3201/eid2606.191436>.
- Hinenoya, A., Nagano, K., Okuno, K., Nagita, A., Hatanaka, N., Awasthi, S.P., Yamasaki, S., 2020b. Development of XRM-MacConkey agar selective medium for the isolation of *Escherichia albertii*. *Diagn. Microbiol. Infect. Dis.* 97, 115006. <https://doi.org/10.1016/j.diagmicrobio.2020.115006>.
- Hinenoya, A., Li, X.P., Zeng, X., Sahin, O., Moxley, R.A., Logue, C.M., Gillespie, B., Yamasaki, S., Lin, J., 2021. Isolation and characterization of *Escherichia albertii* in poultry at the pre-harvest level. *Zoonoses Public Health* 68, 213–225. <https://doi.org/10.1111/zph.12812>.
- Hinenoya, A., Awasthi, S.P., Yasuda, N., Nagano, K., Hassan, J., Takehira, K., Hatanaka, N., Saito, S., Watabe, T., Yoshizawa, M., Inoue, H., Yamasaki, S., 2022a. Detection, isolation and molecular characterization of *Escherichia albertii* from wild birds in West Japan. *Jpn. J. Infect. Dis.* 75, 156–163. <https://doi.org/10.7883/yoken.JJID.2021.355>.
- Hinenoya, A., Wang, H., Patrick, E.M., Zeng, X., Cao, L., Li, X.P., Lindsey, R.L., Gillespie, B., He, Q., Yamasaki, S., Lin, J., 2022b. Longitudinal surveillance and comparative characterization of *Escherichia albertii* in wild raccoons in the United States. *Microbiol. Res.* 262, 127109. <https://doi.org/10.1016/j.micres.2022.127109>.
- Hoque, R., Ahmed, S.M., Naher, N., Islam, M.A., Rousham, E.K., Islam, B.Z., Hassan, S., 2020. Tackling antimicrobial resistance in Bangladesh: a scoping review of policy and practice in human, animal and environment sectors. *PLoS One* 15, e0227947. <https://doi.org/10.1371/journal.pone.0227947>.
- Huseby, D.L., Pietsch, F., Brandis, G., Garoff, L., Tegehall, A., Hughes, D., 2017. Mutation supply and relative fitness shape the genotypes of ciprofloxacin-resistant *Escherichia coli*. *Mol. Biol. Evol.* 34, 1029–1039. <https://doi.org/10.1093/molbev/msx052>.
- Huys, G., Cnockaert, M., Janda, J.M., Swings, J., 2003. *Escherichia albertii* sp. nov., a diarrhoeagenic species isolated from stool specimens of Bangladeshi children. *Int. J. Syst. Evol. Microbiol.* 53, 807–810. <https://doi.org/10.1099/ijs.0.02475-0>.
- Hyma, K.E., Lacher, D.W., Nelson, A.M., Bumbaugh, A.C., Janda, J.M., Strockbine, N.A., Young, V.B., Whittam, T.S., 2005. Evolutionary genetics of a new pathogenic *Escherichia* species: *Escherichia albertii* and related *Shigella boydii* strains. *J. Bacteriol.* 187, 619–628. <https://doi.org/10.1128/JB.187.2.619-628.2005>.
- Iguchi, A., Takemura, T., Ogura, Y., Nguyen, T.T.H., Kikuchi, T., Okuno, M., Tokizawa, A., Iwashita, H., Pham, H.Q.A., Doan, T.H., Tran, N.L., Tran, T.L., Nguyen, T.H., Tran, T.H., Pham, T.N.L., Dao, T.D., Vu, T.M.H., Nguyen, T.N., Vu, H., Nguyen, V.T., Vu, T.T.H., Le, T.H., Lai, T.A., Ngo, T.C., Hasebe, F., Nguyen, D.T., Yamashiro, T., 2023. Genomic characterization of endemic diarrhoeagenic *Escherichia coli* and *Escherichia albertii* from infants with diarrhea in Vietnam. *PLoS Negl. Trop. Dis.* 17, e0011259. <https://doi.org/10.1371/journal.pntd.0011259>.



- Imam, T., Gibson, J.S., Gupta, S.D., Foysal, M., Das, S.B., Hoque, M.A., Fournié, G., Henning, J., 2023. Social and cognitive factors influencing commercial chicken farmers' antimicrobial usage in Bangladesh. *Sci. Rep.* 13, 572. <https://doi.org/10.1038/s41598-022-26859-8>.
- Jacoby, G.A., Strahilevitz, J., Hooper, D.C., 2014. Plasmid-mediated quinolone resistance. *Microbiol. Spectr.* 2. <https://doi.org/10.1128/microbiolspec.PLAS-0006-2013>.
- Jolley, K.A., Maiden, M.C.J., 2010. BIGSdb: scalable analysis of bacterial genome variation at the population level. *BMC Bioinformatics* 11, 595. <https://doi.org/10.1186/1471-2105-11-595>.
- Juraschek, K., Käsbohrer, A., Malorny, B., Schwarz, S., Meemken, D., Hammerl, J.A., 2021. Dissection of highly prevalent *qnrS1*-carrying IncX plasmid types in commensal *Escherichia coli* from German food and livestock. *Antibiotics* 10, 1236. <https://doi.org/10.3390/antibiotics10101236>.
- Kashio, H., Konno, T., Takahashi, S., Suzuki, S., Ito, K., Aoki, J., Sakakida, N., Sato, M., Konishi, N., Obata, H., Kumagai, Y., 2020. Identification of multiple *Escherichia albertii* O-genotypes as etiological agents in a foodborne disease outbreak. *Jpn. J. Food Microbiol.* 37, 183–187. <https://doi.org/10.5803/jsfm.37.183>.
- Konno, T., Takahashi, S., Suzuki, S., Kashio, H., Ito, Y., Kumagai, Y., 2021. Distribution of the O-genotypes of *Escherichia albertii* isolated from humans and environmental water in Akita prefecture, Japan. *Jpn. J. Infect. Dis.* 74, 381–384. <https://doi.org/10.7883/yoken.JJID.2020.731>.
- Kozlov, A.M., Darriba, D., Flouri, T., Morel, B., Stamatakis, A., 2019. RAXML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. *Bioinformatics* 35, 4453–4455. <https://doi.org/10.1093/bioinformatics/btz305>.
- Letunic, I., Bork, P., 2024. Interactive Tree of Life (iTOL) v6: recent updates to the phylogenetic tree display and annotation tool. *Nucleic Acids Res.* 52, W78–W82. <https://doi.org/10.1093/nar/gkac268>.
- Li, Q., Wang, H., Xu, Y., Bai, X., Wang, J., Zhang, Z., Liu, X., Miao, Y., Zhang, L., Li, X., Zou, N., Yan, G., Chen, X., Zhang, J., Fu, S., Fan, R., Xu, J., Li, J., Xiong, Y., 2018. Multidrug-resistant *Escherichia albertii*: co-occurrence of beta-lactamase and MCR-1 encoding genes. *Front. Microbiol.* 9, 258. <https://doi.org/10.3389/fmicb.2018.00258>.
- Lindsey, R.L., Fedorka-Cray, P.J., Abley, M., Turpin, J.B., Meinersmann, R.J., 2015. Evaluating the occurrence of *Escherichia albertii* in chicken carcass rinses by PCR, Vitek analysis, and sequencing of the *rpoB* gene. *Appl. Environ. Microbiol.* 81, 1727–1734. <https://doi.org/10.1128/AEM.03681-14>.
- Lindsey, R.L., Garcia-Toledo, L., Fasulo, D., Gladney, L.M., Strockbine, N., 2017. Multiplex polymerase chain reaction for identification of *Escherichia coli*, *Escherichia albertii* and *Escherichia fergusonii*. *J. Microbiol. Methods* 140, 1–4. <https://doi.org/10.1016/j.mimet.2017.06.005>.
- Liu, B., Zheng, D., Zhou, S., Chen, L., Yang, J., 2022a. VFDB 2022: a general classification scheme for bacterial virulence factors. *Nucleic Acids Res.* 50, D912–D917. <https://doi.org/10.1093/nar/gkab1107>.
- Liu, Q., Bai, X., Yang, X., Fan, G., Wu, K., Song, W., Sun, H., Chen, S., Chen, H., Xiong, Y., 2018. Identification and genomic characterization of *Escherichia albertii* in migratory birds from Poyang Lake, China. *Pathogens* 12, 9. <https://doi.org/10.3390/pathogens12010009>.
- Maeda, E., Murakami, K., Sera, N., Ito, K., Fujimoto, S., 2015. Detection of *Escherichia albertii* from chicken meat and giblets. *J. Vet. Med. Sci.* 77, 871–873. <https://doi.org/10.1292/jvms.14-0640>.
- Magiorakos, A.P., Srinivasan, A., Carey, R.B., Carmeli, Y., Falagas, M.E., Giske, C.G., Harbarth, S., Hindler, J.F., Kahlmeter, G., Olsson-Liljequist, B., Paterson, D.L., Rice, L.B., Stelling, J., Struelens, M.J., Vatopoulos, A., Weber, J.T., Monnet, D.L., 2012. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clin. Microbiol. Infect.* 18, 268–281. <https://doi.org/10.1111/j.1469-0691.2011.03570.x>.
- Masuda, K., Ooka, T., Akita, H., Hiratsuka, T., Takao, S., Fukada, M., Inoue, K., Honda, M., Toda, J., Sugitani, W., Narimatsu, H., Ishioka, T., Hirai, S., Sekizuka, T., Kuroda, M., Morita, Y., Hayashi, T., Kimura, H., Oishi, K., Ohnishi, M., Fujimoto, S., Murakami, K., 2020. Epidemiological aspects of *Escherichia albertii* outbreaks in Japan and genetic characteristics of the causative pathogens. *Foodborne Pathog. Dis.* 17, 144–150. <https://doi.org/10.1089/fpd.2019.2654>.
- Murakami, K., Etoh, Y., Tanaka, E., Ichihara, S., Horikawa, K., Kawano, K., Ooka, T., Kawamura, Y., Ito, K., 2014. Shiga toxin 2f-producing *Escherichia albertii* from a symptomatic human. *Jpn. J. Infect. Dis.* 67, 204–208. <https://doi.org/10.7883/yoken.67.204>.
- Naka, A., Hinenoya, A., Awasthi, S.P., Yamasaki, S., 2022. Isolation and characterization of *Escherichia albertii* from wild and safeguarded animals in Okayama Prefecture and its prefectural borders, Japan. *J. Vet. Med. Sci.* 84, 1299–1306. <https://doi.org/10.1292/jvms.22-0213>.
- Naka, A., Hayashi, K., Ikemoto, S., Ogata, M., 2024. Native wild mammals as *Escherichia albertii* carriers in the Chugoku and Kinki regions of western Japan. *Jpn. J. Vet. Res.* 72, 13–27. <https://doi.org/10.57494/jivr.72.1.13>.
- Nakae, K., Ooka, T., Murakami, K., Hara-Kudo, Y., Imuta, N., Gotoh, Y., Ogura, Y., Hayashi, T., Okamoto, Y., Nishi, J., 2021. Diversification of *Escherichia albertii* H-antigens and development of H-genotyping PCR. *Front. Microbiol.* 12, 737979. <https://doi.org/10.3389/fmicb.2021.737979>.
- Oaks, J.L., Besser, T.E., Walk, S.T., Gordon, D.M., Beckmen, K.B., Burek, K.A., Haldorsen, G.J., Bradway, D.S., Ouellette, L., Rurangirwa, F.R., Davis, M.A., Dobbin, G., Whittam, T.S., 2010. *Escherichia albertii* in wild and domestic birds. *Emerg. Infect. Dis.* 16, 638–646. <https://doi.org/10.3201/eid1604.090695>.
- Ooka, T., Seto, K., Kawano, K., Kobayashi, H., Etoh, Y., Ichihara, S., Kaneko, A., Isobe, J., Yamaguchi, K., Horikawa, K., Gomes, T.A., Linden, A., Bardiau, M., Mainil, J.G., Beutin, L., Ogura, Y., Hayashi, T., 2012. Clinical significance of *Escherichia albertii*. *Emerg. Infect. Dis.* 18, 488–492. <https://doi.org/10.3201/eid1803.111401>.
- Ooka, T., Tokuoka, E., Furukawa, M., Nagamura, T., Ogura, Y., Arisawa, K., Harada, S., Hayashi, T., 2013. Human gastroenteritis outbreak associated with *Escherichia albertii*, Japan. *Emerg. Infect. Dis.* 19, 144–146. <https://doi.org/10.3201/eid1901.120646>.
- Ooka, T., Seto, K., Ogura, Y., Nakamura, K., Iguchi, A., Gotoh, Y., Honda, M., Etoh, Y., Ikeda, T., Sugitani, W., Konno, T., Kawano, K., Imuta, N., Yoshiie, K., Hara-Kudo, Y., Murakami, K., Hayashi, T., Nishi, J., 2019. O-antigen biosynthesis gene clusters of *Escherichia albertii*: their diversity and similarity to *Escherichia coli* gene clusters and the development of an O-genotyping method. *Microb. Genom.* 5. <https://doi.org/10.1099/mgen.0.000314>.
- Page, A.J., Cummins, C.A., Hunt, M., Wong, V.K., Reuter, S., Holden, M.T.G., Fookes, M., Falush, D., Keane, J.A., Parkhill, J., 2015. Roary: rapid large-scale prokaryote pan genome analysis. *Bioinformatics* 31, 3691–3693. <https://doi.org/10.1093/bioinformatics/btv421>.
- Page, A.J., Taylor, B., Delaney, A.J., Soares, J., Seemann, T., Keane, J.A., Harris, S.R., 2016. SNP-sites: rapid efficient extraction of SNPs from multi-FASTA alignments. *Microb. Genom.* 2, e000056. <https://doi.org/10.1099/mgen.0.000056>.
- Parvin, M.S., Ali, M.Y., Mandal, A.K., Talukder, S., Islam, M.T., 2022. Sink survey to investigate multidrug resistance pattern of common foodborne bacteria from wholesale chicken markets in Dhaka city of Bangladesh. *Sci. Rep.* 12, 10818. <https://doi.org/10.1038/s41598-022-14883-7>.
- Sáenz, Y., Zarazaga, M., Brías, L., Ruiz-Larrea, F., Torres, C., 2003. Mutations in *gyrA* and *parC* genes in nalidixic acid-resistant *Escherichia coli* strains from food products, humans and animals. *J. Antimicrob. Chemother.* 51, 1001–1005. <https://doi.org/10.1093/jac/dkg168>.
- Samad, M.A., Ebersson, L., Begum, R., Alam, M.G.S., Talukdar, F., Akter, R., Dang-Xuan, S., Sharma, G., Islam, S., Siddiky, N.A., Uddin, A.A., Asheak Mahmud, M., Sarker, M.S., Rahman, M.S., Grace, D., Lindahl, J.F., 2023. Microbial contamination and antibiotic resistance in marketed food in Bangladesh: current situation and possible improvements. *Antibiotics* 12, 555. <https://doi.org/10.3390/antibiotics12030555>.
- Scuron, M.D., Boesze-Battaglia, K., Dlakić, M., Shenker, B.J., 2016. The cytolethal distending toxin contributes to microbial virulence and disease pathogenesis by acting as a tri-perditious toxin. *Front. Cell. Infect. Microbiol.* 6, 168. <https://doi.org/10.3389/fcimb.2016.00168>.
- Souvorov, A., Agarwala, R., Lipman, D.J., 2018. SKESA: strategic k-mer extension for scrupulous assemblies. *Genome Biol.* 19, 153. <https://doi.org/10.1186/s13059-018-1540-z>.
- Tenover, F.C., Arbeit, R.D., Goering, R.V., Mickelsen, P.A., Murray, B.E., Persing, D.H., Swaminathan, B., 1995. Interpreting chromosomal DNA restriction patterns produced by pulsed-field gel electrophoresis: criteria for bacterial strain typing. *J. Clin. Microbiol.* 33, 2233–2239. <https://doi.org/10.1128/jcm.33.9.2233-2239.1995>.
- Tokoi, Y., Kataoka, S., Wakatsuki, A., Tanizawa, A., Nakada, Y., Seki, A., Ishioka, M., Arai, K., Okago, Y., Kaneko, J., Hase, M., Kihara, H., 2018. Case report of *Escherichia albertii* food poisoning in Utsunomiya city. *Jpn. J. Food Microbiol.* 35, 159–162. <https://doi.org/10.5803/jsfm.35.159>.
- Uddin, J., Hossain, K., Hossain, S., Saha, K., Jubyda, F.T., Haque, R., Billah, B., Talukder, A.A., Parvez, A.K., Dey, S.K., 2019. Bacteriological assessments of foodborne pathogens in poultry meat at different super shops in Dhaka, Bangladesh. *Ital. J. Food. Saf.* 8, 6720. <https://doi.org/10.4081/ijfs.2019.6720>.
- Versalovic, J., Koeuth, T., Lupski, J.R., 1991. Distribution of repetitive DNA sequences in eubacteria and application to fingerprinting of bacterial genomes. *Nucleic Acids Res.* 19, 6823–6831. <https://doi.org/10.1093/nar/19.24.6823>.
- Wang, H., Li, Q., Bai, X., Xu, Y., Zhao, A., Sun, H., Deng, J., Xiao, B., Liu, X., Sun, S., Zhou, Y., Wang, B., Fan, Z., Chen, X., Zhang, Z., Xu, J., Xiong, Y., 2016. Prevalence of *eae*-positive, lactose non-fermenting *Escherichia albertii* from retail raw meat in China. *Epidemiol. Infect.* 144, 45–52. <https://doi.org/10.1017/S0950268815001120>.
- Wang, H., Zhang, L., Cao, L., Zeng, X., Gillespie, B., Lin, J., 2022. Isolation and characterization of *Escherichia albertii* originated from the broiler farms in Mississippi and Alabama. *Vet. Microbiol.* 267, 109379. <https://doi.org/10.1016/j.vetmic.2022.109379>.
- Xu, B., Hatanaka, N., Awasthi, S.P., Hinenoya, A., Yamasaki, S., 2023. Seasonality of detection rate of *Escherichia albertii* in wild raccoons (*Procyon lotor*) in Osaka, Japan. *J. Vet. Med. Sci.* 86, 180–183. <https://doi.org/10.1292/jvms.23-0372>.